

Praneeth Reddy Mallupalli

+1 (224) 536-1466 | praneeth.mallupalli@gmail.com | Evanston, IL | [linkedin.com/in/praneethreddym](https://www.linkedin.com/in/praneethreddym) | github.com/prm036

EDUCATION

Northwestern University, Evanston, IL <i>M.S. in Computer Science (Concentration: Artificial Intelligence)</i> • Coursework: Machine Learning, Deep Learning, Generative AI, Agent AI, NLP, Large Language Models.	Dec 2026(Expected) <i>GPA: 3.9/4.0</i>
Osmania University, Hyderabad, India <i>B.E. in Computer Science and Engineering</i>	June 2021 <i>GPA: 3.9/4.0</i>

TECHNICAL SKILLS

Languages	: Python, C/C++, Java, SQL
ML Frameworks	: PyTorch, Keras, Scikit-learn, NumPy, SciPy, Pandas
Generative AI	: LLMs (Fine-Tuning, Evaluation, Agents), Transformers, Diffusion Models, Quantization
Data & Infrastructure	: AWS, Docker, Kubernetes, Git, FastAPI, ETL Pipelines, PostgreSQL, Distributed Systems

ACADEMIC PROJECTS

- Fine-tuning LLM for Math Problem Solving** | [GitHub](#)
- Fine-tuned Qwen2.5-7B-Instruct model on GSM8K dataset using LoRA, improving the math reasoning by 50%.
 - Mitigated catastrophic forgetting during domain adaptation, while maintaining its safety alignment score of 0.88.
- Post-Training Quantization (PTQ) for Diffusion Transformers: An Empirical Evaluation** | [GitHub](#)
- Benchmarked low-bit FP and INT quantization on PixArt-Alpha (0.6B) diffusion model under equivalent precision budgets.
 - Observed that FP outperformed INT quantization under W4A8, reducing FID from 42.4 to 38.8 while preserving comparable CLIP Score and ImageReward scores.
- LeHome Challenge: VLA Model for Garment Manipulation tasks** | [GitHub](#)
- Fine-tuned **SmolVLA** on teleoperated demonstrations using imitation learning for long-horizon control tasks.
 - Achieved **69%** success rate on garment folding, outperforming Diffusion Policy (**41%**) and ACT (**61%**) baselines.
 - Enhanced model generalization capabilities by augmenting training data with NVIDIA Cosmos-Transfer synthetic data.

PROFESSIONAL EXPERIENCE

- Kilwa Technologies LLC** Chicago, IL
Machine Learning Intern June 2026–Aug 2026
- Designed **METI** (Market Entry Timing Indicator), a hybrid time-series forecasting model that identifies optimal investment windows in **25+** African markets with confidence scores over a **1–6 month horizon**.
 - Factored in Bayesian-optimized sentiment signals, exogenous variables, trend, seasonality, and Monte Carlo simulations to model evolving market conditions and forecast uncertainty.
 - Benchmarked the model against ARIMA+ and zero-shot **TimesFM** baselines, achieving a **15%** lower RMSE while improving the 3-month forecast hit rate from **33%** to over **65%**.
 - Enhanced **ISI** (Investment Suitability Index) module by integrating a hierarchical **XGBoost** ensemble model achieving an AUC of **0.80** for FDI-surge forecasting.
- Arcesium (D.E. Shaw Group)** Hyderabad, India
Senior Software Engineer July 2023 – Aug 2025
- Rearchitected the executor component of a distributed job execution system (processing 1M+ jobs/day) that reduced DynamoDB RCUs by **70%**, removing a key bottleneck for horizontal autoscaling.
 - Improved the security of an enterprise alerting platform by implementing HMAC-based API authentication, while maintaining **sub-0.5 ms** verification latency.
- Software Engineer* June 2021 – June 2023
- Engineered a distributed log ingestion pipeline using the Elastic Stack (ELK), processing over 40 TB/day of streaming logs across EC2 and Kubernetes workloads to support real-time search and analytics.
 - Identified and reported issues observed in **FluentBit** under production workloads, contributing to open-source community.
- Data Science Intern* Feb 2021 – May 2021
- Developed a **statistical model** to quantify the health of core infrastructure services by combining diverse operational telemetry data into a unified, interpretable health score.
 - Applied exponential smoothing (ETS) **time-series models** for trend forecasting and anomaly detection, improving health score accuracy by **30%** over heuristic baseline methods.

LEADERSHIP & HONORS

- Teaching Assistant, Northwestern University — CS496: Machine Learning and Sensing.
- 1st Place, Arcesium Hackathon (2024) themed “Data Science & Analytics at Scale” — Won among 130+ teams.